AI-Generated Song Detection & Attribution

Technical Report

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Task	Pairwise similarity metric for AI-generated music attribution
Approach	Chunk-based feature extraction with weighted similarity scoring
Datasets	MIPPIA SMP, SONICS (97k+), FakeMusicCaps (55k clips)
Key result	Key result Consistency-aware attribution and hybrid AI detection improved separation between related and unrelated demo pairs

1. Problem Statement

The goal is to design a system that ingests two full-length audio tracks and outputs:

- **Attribution Score (0–1)** — the likelihood that Track B is derived from Track A (e.g., an AI-generated cover, plagiarised remix, or hallucinated generation).
- **AI Detection Score (0–1)** — the likelihood that a given track is AI-generated (produced by systems like Suno or Udio).

This goes beyond binary classification. The attribution task requires **relational modelling** robust to tempo variations, arrangement changes, and spectral artifacts introduced by modern AI generators. The detection task requires distinguishing subtle statistical differences between human recordings and machine-generated audio.

2. Solution Architecture

The system follows a four-stage pipeline: audio loading, chunked feature extraction, pairwise similarity computation, and score interpretation. Both tracks are processed identically through the first two stages before being compared in the similarity engine.

Pipeline overview

Stage	Component	Description
1. Ingestion	librosa.load()	Load audio, resample to 22,050 Hz mono
2. Chunking	Overlapping windows	30s windows with 10s hop — covers entire track
3. Extraction	Feature extractor	Per-chunk: MFCCs, Mel, Chroma, Tempo, HNR, Phase, CLAP*
4a. Attribution	Similarity engine	Bidirectional chunk matching + weighted similarity + global consistency factor (order, coverage, mutuality)
4b. AI Detection	Hybrid detector	Logistic Regression score combined with heuristic artifact guardrails

**CLAP embeddings are optional — enabled via --clap flag. Requires ~1 GB model download.*

The attribution score is no longer based only on local best-match chunk similarity. The updated scorer also measures whether the match is globally consistent across the full track, reducing false positives on unrelated music that happens to share local timbral or harmonic similarities.

3. Feature Engineering

Feature selection rationale

Feature selection balances three concerns: (a) capturing timbral and melodic similarity for attribution, (b) detecting AI-specific spectral artifacts, and (c) keeping extraction fast enough for full-length tracks.

Feature	Dim	Rationale
MFCCs (mean + std)	80	Timbral texture — primary fingerprint of instrument/voice identity
Log-Mel Spectrogram	128	Frequency content summary across the full spectrum
Chroma (mean)	12	Pitch-class profile — captures melody and harmonic structure
Tempo (BPM)	1	Rhythmic matching; AI covers often preserve the original tempo
HNR	1	Harmonic-to-Noise Ratio: AI generators produce characteristically extreme values
Spectral Flatness	1	Near-zero in AI output (overly tonal); higher variance in human recordings
Phase Discontinuity	1	AI vocoders introduce unnatural phase jumps in the STFT domain
CLAP embedding*	512	Semantic-level audio representation; strongest single attribution signal

Are standard features sufficient?

Standard features (MFCCs, chroma) capture timbre and harmony but miss AI-specific artifacts. The inclusion of HNR, spectral flatness, and phase discontinuity addresses this gap. Empirical analysis on MIPPIA (human) vs SONICS/FakeMusicCaps (AI) confirmed that **spectral_flatness is the strongest single discriminator** between human and AI audio (Logistic Regression coefficient: −3.64).

Attribution feature weights

Feature group	Weight	Role
CLAP embedding	0.35	Semantic similarity — dominant signal when available
MFCCs	0.15	Timbre fingerprint
Chroma	0.15	Melody and harmony
Log-Mel	0.10	Frequency content
HNR	0.08	AI artifact indicator
Spectral Flatness	0.07	AI artifact indicator
Tempo	0.05	Rhythmic alignment
Phase Discontinuity	0.05	AI artifact indicator

When CLAP is unavailable, weights are renormalised across the remaining features.

4. Handling Variable-Length Audio & Window-Bias Prevention

A naive approach analyses only the first 30 seconds — introducing **window-bias** that favors tracks with similar intros regardless of overall content. This system prevents it through two mechanisms:

- **Overlapping chunking:** Audio is segmented into 30-second windows with a 10-second hop, covering the entire track. A 3-minute track produces ~15 chunks. A final partial chunk is retained if it exceeds 10 seconds.
- **Best-match alignment strategy:** The updated system uses bidirectional chunk matching: chunks from Track A are matched against Track B, and chunks from Track B are matched against Track A. A global consistency factor is then applied on top of the local similarity scores. This factor rewards temporally consistent matches and penalizes accidental local matches that do not preserve global ordering or broad coverage across the destination track.

This approach handles:

- **Overlapping chunking:** 30-second windows with 10-second hop cover the full track.
- **Bidirectional matching:** similarity is measured from A→B and from B→A instead of only one direction.
- **Consistency-aware scoring:** the final attribution score is reduced when the best matches are poorly ordered or weakly distributed across the track.

5. AI Detection — Trained Classifier

Motivation

The current AI detector is hybrid rather than purely classifier-based. It combines a trained Logistic Regression model with feature-based heuristic guardrails derived from spectral artifacts. This was introduced to prevent clearly synthetic tracks from being underestimated by the classifier alone.

Training data

Source	Label	Count
MIPPIA SMP tracks	Human	154
FakeMusicCaps (5 TTM models)	AI	100
SONICS (Suno)	AI	1
Total		**255**

Features used

The final AI score is computed by blending the trained classifier probability with a heuristic score based on spectral_flatness, phase_discontinuity, and HNR. A guardrail is applied when spectral flatness is strongly AI-like but the classifier remains underconfident.

Thresholds:

- >= 0.75: Likely AI-generated
- 0.55–0.75: Possibly AI-generated
- 0.40–0.55: Ambiguous
- < 0.20: Likely human-made

Demo examples:

- Jennifer Rush original: 0.0862
- Céline Dion remake: 0.0532
- SONICS Suno track: 0.7679

Results (5-fold stratified cross-validation)

Metric	Score
Accuracy	0.867 ± 0.049
Precision	0.767 ± 0.060
Recall	**0.960 ± 0.080**
F1	**0.851 ± 0.057**

High recall (0.96) is the most important property for this use case — the system is unlikely to miss an AI-generated track.

Feature importance

Spectral_flatness dominates with a coefficient of −3.64, confirming that AI generators produce abnormally low spectral flatness compared to human recordings. `zcr_mean` and `mfcc1_mean` contribute secondary discriminative power.

Important caveat

The AI detection score is a probabilistic estimate, not a proof. Scores are reported as likelihood values (e.g., “likely AI-generated”) rather than categorical verdicts. The model was trained on 255 samples — a larger, more diverse training set would improve both precision and generalisation.

6. Attribution — Evaluation on MIPPIA SMP

Dataset

The attribution module was initially evaluated on 51 complete downloadable pairs from the MIPPIA SMP dataset, covering the relation types plag, plag_doubt, and remake. However, those benchmark values were obtained with the earlier attribution scorer, which relied primarily on local best-match chunk similarity.

Under that earlier scorer, the evaluated MIPPIA pairs produced very high attribution scores, with a tight score distribution across the related pairs. However, additional testing on unrelated examples revealed that the earlier scoring strategy could inflate similarity for tracks that happened to share local timbral or harmonic patterns without being true derivatives of one another.

To address this, the attribution module was recalibrated. The current scorer still operates on chunk-level audio features, but now combines bidirectional chunk matching with a global consistency factor. In addition to local chunk similarity, the updated score considers whether the best matches preserve temporal order and provide broad coverage across the compared track. This change was introduced specifically to reduce false positives on unrelated music.

Relation	Count
plag	21
plag_doubt	16
remake	13

At the current stage, the most reliable summary of the updated scorer is given by direct sanity-check examples:

- Jennifer Rush original ↔ Céline Dion remake: 0.7074
- Jennifer Rush original ↔ SONICS Suno track: 0.3168
- Jennifer Rush original ↔ Summer Dream: 0.2494

These examples show a more realistic separation between related and unrelated pairs than the earlier scorer. In particular, the recalibrated system keeps the known remake pair above the decision boundary while pushing unrelated examples substantially lower. Based on these checks, a threshold around 0.55 appears more suitable for the current consistency-aware attribution model.

7. Datasets

Three datasets were used across the two tasks. The **MIPPIA SMP** dataset (158 labeled pairs: remake, plag, plag_doubt) provided ground-truth for attribution evaluation and human-track feature profiling. **SONICS** (97k+ Suno/Udio tracks) and **FakeMusicCaps** (55k clips from five text-to-music models: AudioLDM2, MusicGen, MusicLDM, Mustango, Stable Audio Open) supplied the AI-generated examples for classifier training and artifact analysis. All datasets are excluded from the repository; download instructions are provided in README.md.

8. Tools & Libraries

Category	Library	Purpose
Audio I/O & DSP	librosa	Loading, resampling, STFT, MFCCs, chroma, beat tracking
Audio I/O	soundfile	WAV/FLAC reading
Neural embeddings	transformers	CLAP model (laion/clap-htsat-unfused)
Deep learning	torch	CLAP inference backend
ML classifier	scikit-learn	Logistic Regression, StandardScaler, cross-validation
Numerics	numpy, scipy	Array operations, signal processing
Data	pandas, datasets	Dataset loading and manipulation
Visualisation	matplotlib, seaborn	Feature distribution plots
Dataset download	yt-dlp	MIPPIA audio retrieval from YouTube

9. Design Decisions & Trade-offs

Decision	Rationale	Trade-off
30s chunks, 10s hop	Covers full track; captures local patterns while maintaining global coverage	Higher compute cost vs. a single global embedding approach
Best-match (not DTW)	Simple, fast, handles segment reordering naturally	May overestimate similarity for partially similar tracks
Cosine over Euclidean	Scale-invariant; better suited for high-dimensional embeddings	Loses magnitude information
Logistic Regression	Interpretable, calibrated probabilities, fast inference	Less powerful than deep neural classifiers
CLAP as optional	System works on CPU without large downloads	Weaker genre-level attribution without neural embeddings
max_diff per scalar	Domain-adapted normalisation for each scalar metric	Requires domain knowledge to tune correctly

10. Limitations

- **AI detector training set:** 255 samples is a modest set. The classifier may not generalise well to AI generators not represented in FakeMusicCaps/SONICS. Retraining on the full SONICS dataset (97k tracks) would substantially improve robustness.
- **Attribution ≠ AI detection:** A high attribution score indicates the tracks are musically related — it does not imply that either track is AI-generated. These are two separate tasks with separate outputs.
- **CLAP dependency:** The strongest attribution feature requires ~1 GB model download and benefits from GPU acceleration. Without CLAP, the system relies entirely on spectral features.
- **No tempo normalisation:** Chunks are compared at their original BPM. A significantly sped-up or slowed-down cover may produce a lower attribution score.
- **Stereo → mono collapse:** Stereo panning information is discarded during preprocessing.
- **YouTube availability:** MIPPIA audio is sourced from YouTube — availability varies over time, limiting reproducibility.
- **Best-match inflation:** The max-pooling strategy can inflate scores when one strong chunk in B matches many chunks in A. A symmetric matching approach (e.g., Hungarian algorithm) would mitigate this.

11. Future Improvements

- **Larger training set:** Expand the AI detector with a larger and more diverse AI corpus, including more SONICS samples and additional generators such as MusicGen and Stable Audio.
- **Dynamic Time Warping (DTW):** Evaluate DTW or other sequence-alignment methods on top of the current chunk-based similarity matrix for more principled handling of time-stretched covers.
- **MERT embeddings:** Replace or complement CLAP with the music-specific MERT model for richer semantic representations.
- **Tempo normalisation:** Automatically detect BPM ratio between tracks and align chunks before comparison.
- **Explainability:** Add SHAP values or per-feature contribution heatmaps to explain why a particular score was assigned.
- **End-to-end fine-tuning:** Fine-tune CLAP on MIPPIA pairs as a contrastive learning task to learn attribution-specific representations.
- Evaluate the recalibrated scorer on the full MIPPIA set and refresh all benchmark tables

12. Usage

Full pipeline (requires all three datasets):

```
python evaluate_mippia.py        # 1. Attribution on all MIPPIA pairs; builds features/ cache
python train_ai_detector.py      # 2. Train Logistic Regression AI detector (CV F1 = 0.85)
python evaluate_attribution.py   # 3. Compute Precision / Recall / F1
```

Compare any two tracks (no datasets needed):

```
python compare_tracks.py original.mp3 suspected_cover.mp3
python compare_tracks.py original.mp3 suspected_cover.mp3 --clap   # with CLAP embeddings
python compare_tracks.py original.mp3 suspected_cover.mp3 --json   # JSON output
```

Minimum recommended duration: ~40 seconds per track. Without the trained model, AI detection falls back to a heuristic profile.

End-to-end demo:

```
jupyter notebook demo_notebook.ipynb
```

Demonstrates both tasks on a real MIPPIA pair (Jennifer Rush / Céline Dion) and a SONICS AI track.

The **JSON output** of **compare_tracks.py** now includes attribution diagnostics (base similarity, order consistency, coverage ratio, mutual match ratio, and consistency multiplier), which are useful for debugging false positives and explaining score calibration.

If **similarity_engine.py** is modified, both **evaluate_mippia.py** and **evaluate_attribution.py** should be rerun so that reported benchmark numbers remain aligned with the active scorer.